Input Form (IF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *LTZGLUE* | ***Context:*** *Luxembourgish Professional Newsroom*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Modality alignment is good: both benchmark and deployment are text-only [Q17], RTL articles, comments, parliamentary transcripts, and chat rooms provide register diversity in pre-training [Q74], and OpenLID filtering removes non-Luxembourgish content [Q47]. The 40–400 word filter [Q47] is plausibly compatible with newsroom article lengths. However, Luxembourgish has 'ongoing standardisation' and 'vast amounts of variation' [Q7], and the documentation does not describe explicit orthographic normalisation across datasets. The BPE tokenizer (50,368 vocab on 233M tokens) [Q128, Q75] may not adequately represent journalistic orthographic variants or code-switched French/German tokens, particularly given the formal-register skew of the data [Q112]. The deployment explicitly applies the flexible journalistic register [A3], but LA Ortho ground truth is built from Spellchecker.lu data [Q31] encoding the formal ZLS standard [WEB-11, WEB-12], creating a known register mismatch.

ID	Evidence
Q17	“This subset is split equally, with one half serving as the positive class with original headlines, and the other half providing the article bodies for which we assign swapped headlines.” (p.3)
Q74	“The pre-training set is compiled from a variety of sources of LTZ. A large portion of the data stems from RTL (see Section 3), including news articles (News), transcribed radio interviews (Radio), and user comments (Comments). We also include transcribed podcasts (Podcasts) and transcribed political speeches and debates from the Chambre des Députés (Chamber). In addition, we use 1M sentences from the web crawl of the Leipzig Collection (Web, this excludes RTL), text crawled from LTZ chat rooms (Webchat), a Wikipedia crawl from October 2023 (Wikipedia), and finally, example sentences from the LOD retrieved in March 2024.” (p.6)
Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
Q7	“Additional factors, such as the ongoing standardisation of the language (Gilles, 2019), vast amounts of variation (Lutgen et al., 2025), and decentralised resources, make it extremely challenging to evaluate LTZ language understanding in language models.” (p.1)
Q128	“Both models share a GPTNeoXTokenizerFast tokenizer (Black et al., 2022), a BPE-based tokenizer, which we train on the entire pre-training set, using a minimum frequency of two and a vocabulary size of 50,368.” (p.12)
Q75	“We filter out sentences containing fewer than three words (as tokenized by whitespace), totalling 11.7M sentences, which corresponds to roughly 233M tokens using our tokenizer.” (p.6)
Q112	“Coverage across domains, registers, and demographic varieties may also be limited. LTZ displays substantial orthographic and sociolinguistic variation, yet most data sources reflect formal writing or institutional usage and therefore do not fully represent informal and multilingual contexts.” (p.9)
Q31	“The first class interferes with the subject-verb agreement by changing the conjugated form of the main verb or auxiliary verb. The second class similarly modifies the declined form of the adjective and therefore violates the agreement in case, number, and gender. For the third class, we manipulate the syntax by deleting 2-3 random words from the sentence, depending on the length. The last class impacts the orthography, which is achieved by using data provided by Spellchecker.lu, a semiautomatic spellchecking website frequently used in Luxembourg. We changed one random word in the sentence by using the least frequent variant in the spellchecker data.” (p.3)
WEB-11	LA Ortho is anchored in Spellchecker.lu which encodes the formal ZLS standard, structurally divorced from the journalistic register the deployment applies (https://data.public.lu/en/organizations/zenter-fir-dletzebuerger-sprooch/)
WEB-12	https://men.public.lu/en/grands-dossiers/systeme-educatif/promotion-langue-luxembourgeoise.html

Strengths

- Text-only modality matches the deployment's text-only pipeline [Q17]
- Pre-training corpus draws from a broad set of Luxembourgish written sources including RTL news, comments, parliamentary speech, podcasts, chat rooms, and Wikipedia [Q74], providing some register breadth
- Article-length filtering (40–400 words) [Q47] is operationally compatible with newsroom article lengths

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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Both benchmark and deployment use plain text [Q17]; signal-distribution mismatch is low at the modality level. Tokenisation (BPE, 50,368 vocab on 233M tokens) [Q128, Q75] may under-represent journalistic orthographic variants and code-switched tokens.

IF-2: Luxembourg has near-universal digital infrastructure (98.8% internet penetration [WEB-17]), so capture infrastructure is not a constraint.

IF-3: The benchmark filters articles to 40–400 words [Q47] and removes non-Luxembourgish content via OpenLID [Q47] — both operationally plausible for newsroom text. However, no explicit orthographic normalisation across datasets is documented despite acknowledged 'vast amounts of variation' [Q7], and the LA Ortho category encodes the ZLS formal standard via Spellchecker.lu [Q31, WEB-11] which diverges from the deployment's flexible journalistic register [A3].

IF-4: Form-level mismatches that may harm external validity: (a) tokenisation potentially mis-segmenting code-switched and orthographically variant tokens; (b) implicit assumption of formal orthography in LA Ortho ground truth diverging from journalistic practice [A3, Q31].

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Q75	“We filter out sentences containing fewer than three words (as tokenized by whitespace), totalling 11.7M sentences, which corresponds to roughly 233M tokens using our tokenizer.” (p.6)
WEB-17	98.8% internet penetration in Luxembourg confirms enterprise-grade digital infrastructure for newsroom deployment (https://www.theglobaleconomy.com/Luxembourg/Internet_users/)

Q47	“We applied a series of preprocessing steps to ensure data quality. Specifically, we removed articles identified as non-Luxembourgish by OpenLID (Burchell et al., 2023), as well as those containing fewer than 40 words or more than 400 words.” (p.4)
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Information Gaps

- Whether explicit orthographic normalisation was applied across datasets (gap_id 3, partially resolved — formal-source inputs confirmed but normalisation across datasets undocumented)
- Tokenizer behaviour on code-switched and orthographically variant tokens

Requires Expert Verification

- Whether the BPE tokenizer adequately segments code-switched newsroom copy

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Remediation

Gap 1: No explicit orthographic normalisation across datasets despite acknowledged 'vast amounts of variation' [Q7]; tokenizer behaviour on code-switched and orthographically variant tokens is undocumented [Q128]

Recommendation: Profile the BPE tokenizer's handling of code-switched French/German tokens and journalistic orthographic variants on a sample of newsroom copy; consider tokenizer extension or fine-tuning if fragmentation rates are high

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